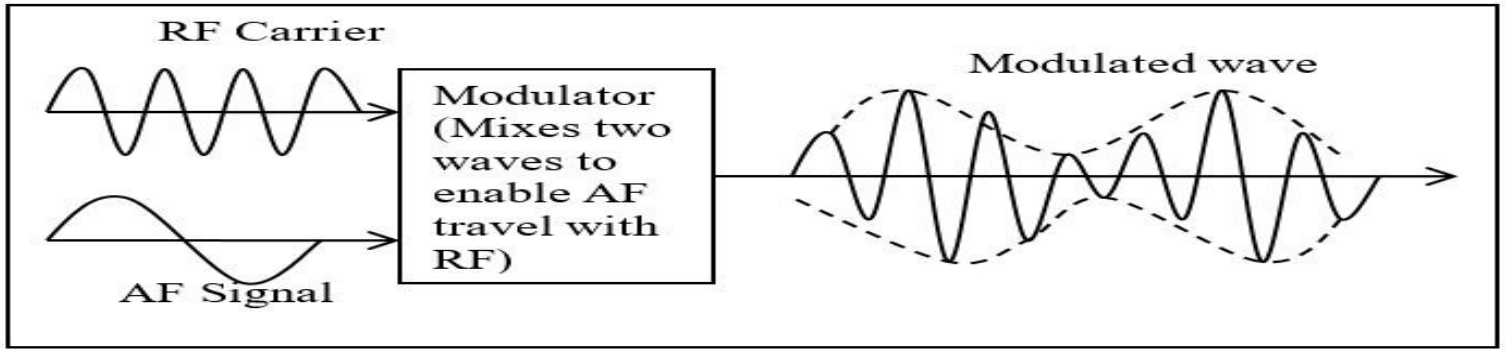
MODULATION

MODULATION

- *Modulation is the process of combining an (low) audio-frequency (AF) signal with a very high-frequency called radio-frequency (RF) to produce a modulated carrier wave* at the transmitting station
- During modulation, some characteristic (amplitude, frequency or phase) of the carrier wave is varied in time with the modulating signal and is accomplished by combining the two signals
- The fluctuations in (amplitude, frequency or phase) of the carrier wave depend on the corresponding characteristic of the audio signal.



Baseband Signal

A baseband signal (audio or video) or any waveform containing information is called a source or information signal. Such signal is not suitable for direct transmission because:

- It has relatively short range.
- Transmitting these low-frequency signals directly results in mutual interference which renders all signals ineffective.
- Size of antennas required for its efficient radiation would be very large.
- It does not lie in a frequency band that can be transmitted over the media.
- It does not fully utilize the capacity of the channel to facilitate signal transmission.

RADIO FREQUENCY(RF)SPECTRUM

Radio frequencies used by different communication systems extend from **very low** frequencies to **extra high** frequencies as tabulated below along with their acronym abbreviations

Frequency	Designation	Abbr.	Uses
3-30kHz	very low frequency	VLF	long distance telegraph broadcasting.
30-300kHz	low frequency	LF	long distance point-to-point services, navigational aids, sound broadcasting and line carrier systems.
0.3-3MHz	Medium frequency	MF	sound broadcasting, ship-shore services and line carrier systems.
3-30MHz	high frequency	HF	medium and long distance point-to-point services, sound broadcasting and linear carrier systems.
30-300MHz	very high frequency	VHF	short distance communication, TV and sound broadcasting and radar.
0.3-3GHz	ultra high frequency	UHF	sound broadcasting and radar.
3-30GHz	super high frequency	SHF	outer-space radio communication.
30-300GHz	extra high frequency	EHF	point-to-point microwave communication and radar.

Benefits of modulation

1. Ease of radiation

If the communication channel consists of free space then aerials will be required to radiate and/or receive the signals. In order to have efficient radiation and reception of signals, the length of the antenna (aerial) would need to be about a quarter wavelength, $\left(\frac{\lambda}{4}\right)$ of the signals being radiated or received. Many signals, audio signals inclusive have frequency components that could be down to **300 MHz** or even lower.

For a frequency of say 1 kHz to be radiated efficiently the aerial will need to be **75 km** long.

$$\lambda = \frac{c}{f} \quad \therefore \quad \frac{3 \times 10^{8}}{1 \times 10^{3}} = 300 \text{ km}$$

Since aerial length requirement is $\frac{\lambda}{4}$ $\therefore$ $\frac{300}{4} = 75$ km. That is how 75 km is arrived at.

This is impractical hence it is difficult to transmit a low frequency signal directly as a radio wave. However, by the process of modulation where the message signal is impressed on a high frequency carrier say 30 MHz, a realistic antenna length is obtained.

$$\lambda = \frac{c}{f} = \frac{3 \times 10^{8}}{30 \times 10^{6}} = 10\text{m}$$

Requirement is $\frac{\lambda}{4} = \frac{10}{4} = 2.5\text{m}$

The antenna need not be more than 2.5 m long.

2. Multiplexing

If more than one signal is to be transmitted over a common channel, modulation could be used to translate different signals to different locations on the frequency spectrum, thus enabling the receiver to select only the desired signal. Several signals being transmitted over a single channel without interference is referred to as multiplexing and the device which facilitates the process is called a multiplexer.

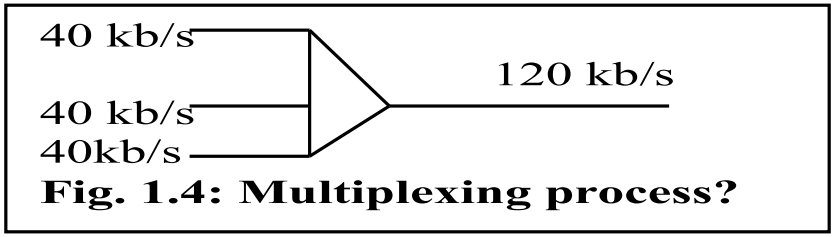


Fig. 1.4: Multiplexing process?

3 Frequency Allocation: Modulation allows several radio or TV signals to broadcast simultaneously at different carrier frequencies allocated to them. The receivers could be tuned to select different stations as desired.

4 To Overcome Equipment Limitation

The performance of signal processing devices such as amplifiers and filters and the case with which they can be built depend very much on the signal frequency. Modulation can be used to translate the signal to a location in the frequency domain where design requirements are easily met.

5 To Reduce Noise and Interference

Some modulation schemes have been found to reduce greatly the effect of noise and interference in communication systems. It is noteworthy here that noise and interference cannot entirely be eliminated in communication systems, but can be minimized.

TYPES OF MODULATION

➢Analog Modulation

- Amplitude Modulation (**AM**)
- Frequency Modulation (**FM**).
- Phase Modulation (**PM**)

➢Digital Modulation

- Minimum Shift Keying (**MSK**).
- Binary Phase Shift Keying (**BPSK**).
- Quadrature Phase Keying (**QPK**)
- And so on

Amplitude Modulation (1)

Amplitude Modulation is the simplest and earliest form.

Amplitude Modulation is a process where the amplitude of a carrier signal is altered according to information in a message signal.

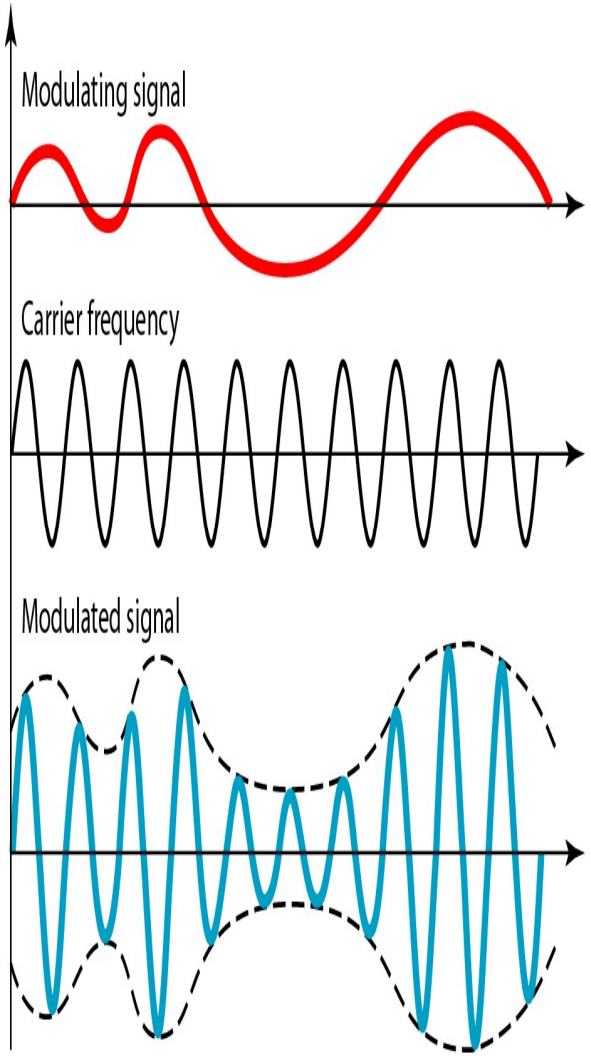
Mathematically, the combination of any two sine waves through a non-linear device produces the following characteristics of a modulator:

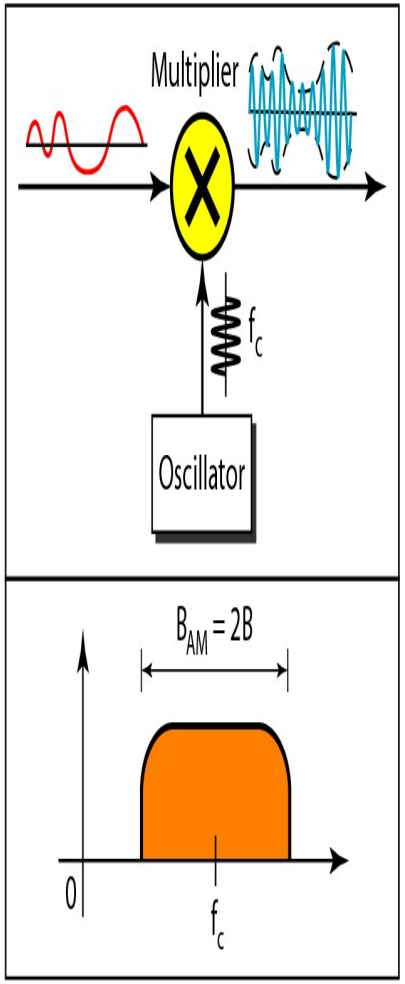
1. Components of DC level
2. Components at each of the two original frequencies.
3. Components of the sum and difference of the frequencies of the two original frequencies.
4. Harmonics of the sum and difference of the frequencies of the two original frequencies.

Mathematical Analysis of Modulation

The message wave and carrier signal are respectively denoted as: $m(t) = A_m \cos w_m t$

$$c(t) = A_c \cos w_c t$$





Amplitude Modulation (2)

The modulated signal is represented by multiplying the carrier signal and the summation of 1 plus the message signal

$$AM(t) = [(1 + A_m \cos w_m t) \times A_c \cos w_c t]$$
$$AM(t) = A_c[\cos w_c t + A_m(\cos w_m t)\,(\cos w_c t)]$$

From trigonometry: $\cos A \cos B = \frac{1}{2}\cos A + B + \frac{1}{2}\cos A - B$

$$[A_c \cos w_c t + \frac{A_m}{2}\,(\cos w_c + w_m)t\,(\cos w_c - w_m)t]$$

but modulation index (m)=$\frac{A_m}{A_c}$

The value **m** gives the extent or depth of modulation because it compares ratios of the amplitude of the message and that of the carrier. However, **m** must be less than one (**m<1**) in order to avoid envelope distortion due to over modulation.

$$[A_c \cos w_c t + \frac{mA_c}{2}\,(\cos w_c + w_m)t + \frac{mA_c}{2}(\cos w_c - w_m)t]$$

Since $w = 2\pi f$ in terms of frequency

then

$$[A_c \cos 2\pi f_c t + \frac{mA_c}{2}\,\cos 2\pi(f_c + f_m)t + \frac{mA_c}{2}\cos 2\pi(f_c - f_m)t]$$

Amplitude Modulation (3)

The three components $f_c + f_m$, $f_c - f_m$, and f_c form AM wave, where:

$f_c + f_m$ is called the upper side band (USB) frequency

$f_c - f_m$ is called the lower side band (LSB) frequency

f_c is carrier wave center frequency

The total bandwidth required for AM can be determined from the bandwidth of the audio signal:

$B_{AM} = 2B$

Amplitude Modulation (4)

➢Advantages of AM

1. Simplest type of modulation technique

2. Hardware design of both transmitter and receiver is very simple and less cost effective

➢Disadvantage of AM

AM is very susceptible to noise

➢Application

1. AM radio broadcast

2. AM is used in shortwave and radio wave broadcasting

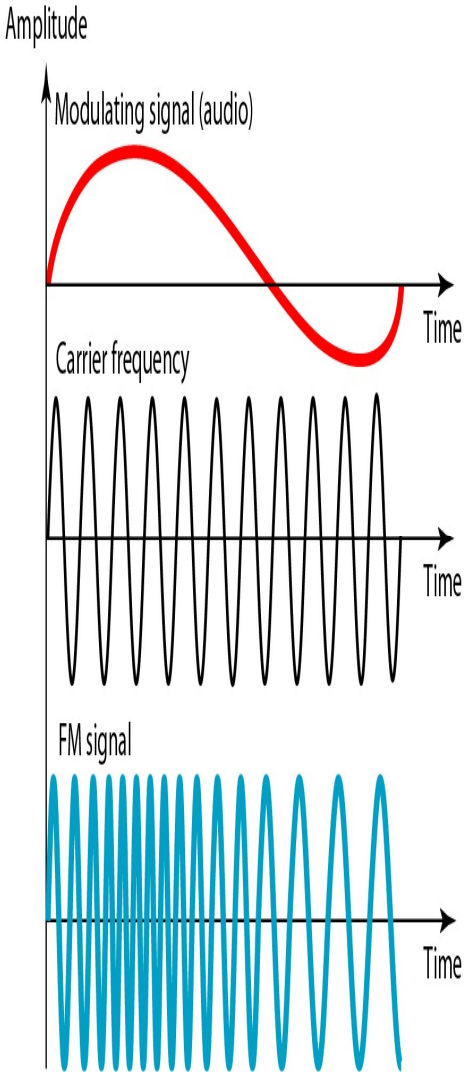
3. AM applications include broadcasting in medium- and high-frequency applications, and aircraft communications

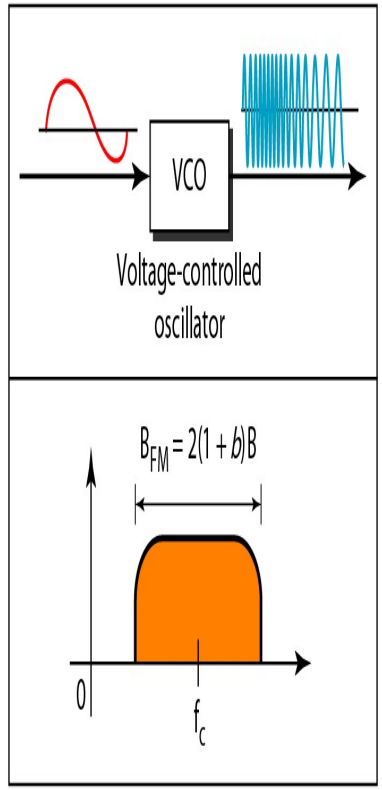
Frequency Modulation (1)

- In this modulation the frequency of the carrier signal is altered in proportion to the message signal while the phase and amplitude are kept constant.

- Frequency modulation is considered to be superior compared to the Amplitude modulation due to better noise immunity and its ability to reject the interfering signals due to the capture effect.
  - ✓ The **capture effect** is defined as the complete suppression of the weaker signal at the receiver's limiter (if present) where the weaker signal is not amplified, but attenuated.

- The total bandwidth required for FM can be determined from the bandwidth of the audio signal: $B_{FM} = 2(1 + \beta)B$
- Modulation index $\beta = \frac{\text{peak frequency deviation}}{\text{modulating frequency}} = \frac{A_m K_w}{f_m}$
- A_m and f_m respectively represent the maximum amplitude and frequency of the modulating signal (information signal). K_w is a positive constant called the frequency sensitivity.





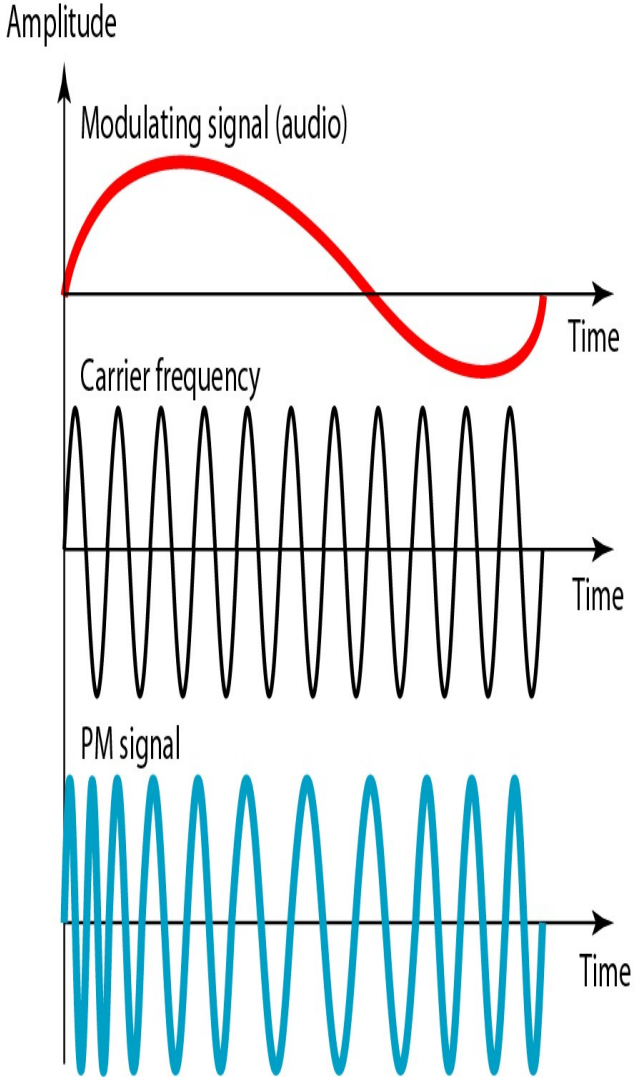
Frequency Modulation (2)

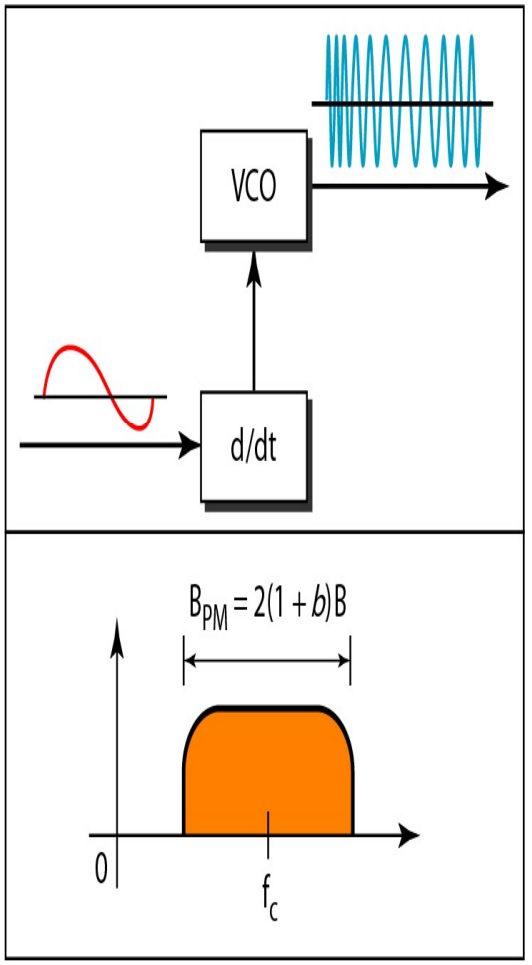
- Advantages:
  - ✓ Amplitude of the frequency modulated wave remains unaffected.
  - ✓ Large decrease in noise, hence increase in S/N ratio.
  - ✓ Noise may reduce by increasing deviation
  - ✓ Frequency allocation allows for a guard band which reduces adjacent channel interference.
  - ✓ FM permits several independent transmitters on the same frequency with negligible interference
- Disadvantages
  - ✓ FM wave can't cover large area.
  - ✓ Transmitting & receiving equipment for FM are complex & costly.
  - ✓ Reception using conventional method is limited to line of sight.
- Applications
  - ✓ Magnetic tape storage: FM is used at intermediate frequencies by analog VCR systems (including VHS) to record both the luminance(black and white) portions of the video signal
  - ✓ Sound : FM is also used at audio frequencies to synthesize sound, which is known as FM synthesis.
  - ✓ Radio : FM is used in many radio stations (FM Radio Stations) all over the world. Normally the frequency range of FM channels are between 88.0 MHz - 108.0 MHz.

Phase Modulation (1)

- This is the modulation where the phase of the carrier signal is altered according to the low frequency of the message signal.

- The total bandwidth required for PM can be determined from the bandwidth
and maximum amplitude of the modulating signal:
$B_{PM} = 2(1 + \beta)B$.





Phase Modulation (2)

➢Advantage

Modulation and demodulation does not catch any channel noise

➢Disadvantage

Circuit needed for PM modulation and demodulation is bit complicated than AM and FM

➢Application

1. Pulse modulation is known for digital signal modulation
2. Satellite communication

Video Demo


08:08
Friday, February 24
Notepad
Themes
Google
Play Store
TP-LINK
FM 88.05MHz
JBL

Video Demo

